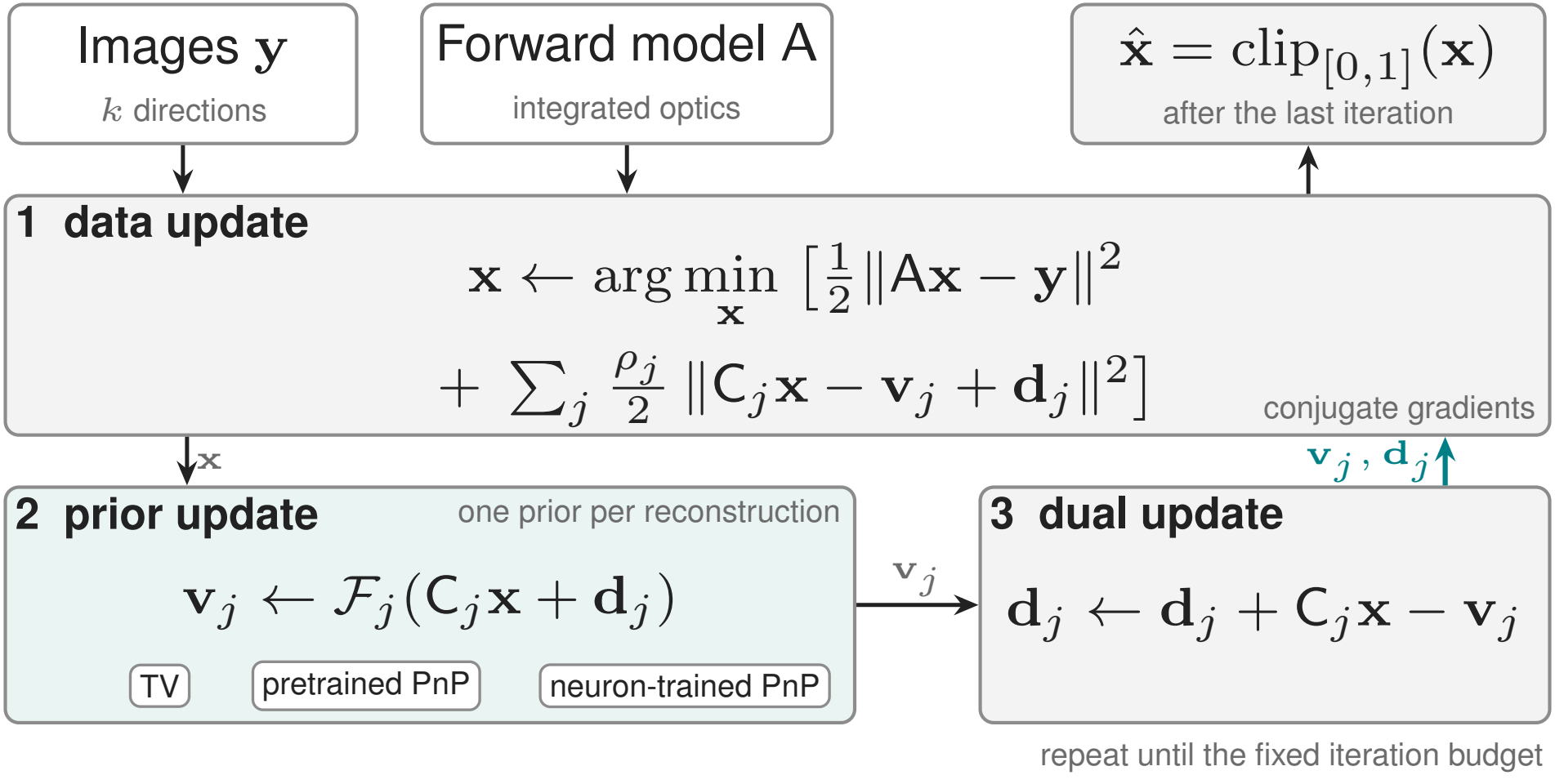

Images $\mathbf{y}$
k directions
Forward model A
integrated optics
$\hat{\mathbf{x}} = \mathrm{clip}_{[0,1]}(\mathbf{x})$
after the last iteration
1 data update
$\mathbf{x} \leftarrow \arg\min_{\mathbf{x}} \left[\frac{1}{2} \|\mathsf{A}\mathbf{x} - \mathbf{y}\|^2 + \sum_j \frac{\rho_j}{2} \|\mathsf{C}_j\mathbf{x} - \mathbf{v}_j + \mathbf{d}_j\|^2 \right]$
conjugate gradients
$\mathbf{x}$
$\mathbf{v}_j$, $\mathbf{d}_j$
2 prior update
one prior per reconstruction
$\mathbf{v}_j \leftarrow \mathcal{F}_j(\mathsf{C}_j\mathbf{x} + \mathbf{d}_j)$
TV
pretrained PnP
neuron-trained PnP
$\mathbf{v}_j$
3 dual update
$\mathbf{d}_j \leftarrow \mathbf{d}_j + \mathsf{C}_j\mathbf{x} - \mathbf{v}_j$
repeat until the fixed iteration budget